Project LEE
Implementation Specification

UI Simplification + CIL Model-Routing Authority

Execution document for Replit - create tasks and implement, do not merely review

Execution Directive

This is an implementation specification. Do not only read, summarize, or comment on it. Create a task plan in the Project LEE task system and then execute the tasks.

Required behavior:

- Create separate implementation tasks for the CIL authority work, the desktop UI consolidation, the Android/mobile consolidation, and final validation.
- Break large workstreams into smaller tasks when needed so each task can be executed, tested, and reviewed independently.
- Preserve all working functionality, data, APIs, desktop packaging, Android behavior, and current integrations.
- Do not delete internal capabilities merely because they are removed from normal navigation.
- Do not claim completion until runtime behavior, typechecks, builds, and relevant tests pass.

Recommended task sequence:

1. Enforce CIL as the mandatory pre-model authority and remove duplicate LEE routing authority.
2. Integrate the read-only CIL model inventory for visibility and economics.
3. Consolidate the desktop Console navigation and contextualize internal capabilities.
4. Simplify the Android companion navigation while preserving local-first capture and approvals.
5. Run regression, build, routing, and UX validation; report all remaining limitations.

Do not create one giant untestable task if the work is better split into multiple executable tasks.

Part A - CIL Must Be the Pre-Model Authority

LEE is already connected to CIL. Do not rebuild the existing signed connection. Harden the authority boundary around that working integration.

A1. Mandatory CIL-before-model flow

For every normal LEE-originated AI/reasoning request, enforce:

```
Identity -> Constitution -> Intent -> Query Engine -> Context Economy -> CIL API
```

CIL then owns the resolution decision:

- T1_TRIGRAM - return the governed CIL answer directly; no model call.
- T2_SEMANTIC - return the governed CIL answer directly; no model call.
- T3_FRONTIER - CIL returns the approved model/provider/route; only then may LEE execute a model call.

No normal LEE path may choose or call an external/paid model before CIL evaluates the request.

A2. Remove duplicate CIL routing authority from LEE

LEE may keep the CIL API client, schemas, auth/signing, health state, correlation IDs, normalized events, telemetry, and UI status. LEE must not independently own:

- T1/T2/T3 resolution decisions
- cognitive-asset selection
- model selection or model-class selection
- provider selection
- cost-based model-routing policy
- CIL-owned contradiction/drift arbitration
- fallback model selection after provider failure

A3. Model Router is execution-only

The Model Router executes the route CIL selected. It may handle transport, endpoint selection, request formatting, response normalization, telemetry, usage accounting, and safe retries.

It must not silently replace CIL's model or provider decision.

A4. Model execution failure returns to CIL

If a CIL-selected model route fails:

```
Model execution failure -> LEE -> CIL reroute request -> new CIL-owned route -> Model Router
executes new route
```

LEE must not locally choose a replacement model.

A5. Replit AI Bridge

Preserve the long-term desktop path:

```
LEE Core on K6 -> CIL -> CIL-selected route -> Model Router -> Replit AI Bridge -> Replit-
managed model -> LEE
```

The Replit AI Bridge is an execution gateway that preserves Replit-managed AI access without requiring separate provider API keys on the K6. It does not choose models.

A6. CIL model inventory is visibility-only

Integrate the authenticated CIL model-inventory endpoint when available. LEE may display:

- total configured, enabled, available, and unavailable models
- providers
- model IDs and route IDs
- availability/health status

The inventory does not grant LEE model-selection authority. Normal execution must still use the model route returned by CIL.

A7. Prove CIL economics

System Economics should measure enough data to demonstrate CIL's actual effect on model spend, including:

- T1 and T2 reuse counts/rates
- T3 escalation counts/rates
- models/providers selected by CIL
- frontier calls avoided
- measured model cost
- estimated equivalent cost without reuse when defensible
- measured/estimated savings with clear labeling
- latency and failure/reroute counts

Do not present estimated savings as measured facts.

Part B - UI Simplification and Consolidation

Product principle:

SIMPLE ON THE OUTSIDE. A MONSTER UNDER THE HOOD.

The current Console exposes too much of LEE's internal architecture directly. Keep the engines, ledgers, diagnostics, and deep controls, but organize normal navigation around owner goals rather than subsystem names.

B1. Primary desktop navigation

```
TODAY
ASK LEE
WORK
  - Projects
  - Portfolio
  - People
  - Objectives
KNOWLEDGE
SYSTEMS
SETTINGS
```

Avoid additional top-level destinations unless there is a strong user-facing reason.

B2. Today

Today should answer: What deserves my attention? Organize around:

- NEEDS YOU - approvals, decisions, blocked/urgent items, consequential holds.
- IN MOTION - active projects, active workflows, meaningful momentum.
- WATCHING - waiting, risks, drift, stale dependencies, degraded systems.
- LEE SUMMARY - what changed, what matters, why, and useful next actions.

Quiet when healthy. Prominent when actionable.

Healthy confidence, CIL, CerbaSeal, backups, connectors, economics, and health indicators should not dominate Today unless they need attention.

B3. Ask Lee

Ask Lee should be the universal deep-access interface. The user should be able to ask about assumptions, decisions, evidence, risks, waiting, project history, system health, CIL savings, CerbaSeal holds, momentum, lessons, and other information already known by LEE without hunting through technical pages.

B4. Work

Work contains Projects, Portfolio, People, and Objectives. Each project should open into one consolidated workspace with:

- Overview - state, momentum, readiness, risks, next actions.
- Activity - timeline, events, meaningful changes.
- Knowledge - Facts, Interpretations, Assumptions, Decisions, Evidence, Strategic Anchors.
- People - relationships, commitments, waiting loops.
- Systems - repos, services, dependencies, health.
- Strategy - recommendations, opportunities, simulations, relevant reflections.

B5. Knowledge

Consolidate Evidence, Facts, Interpretations, Assumptions, Decisions, Decision Patterns, Strategic Anchors, Timeline, Institutional Knowledge, Knowledge Map, and related learning/history into one Knowledge area using tabs, filters, expandable sections, and contextual detail.

B6. Systems

Consolidate connected Lamont Labs systems, CIL, CerbaSeal, Replit AI Bridge, connectors, APIs, health, governance state, approvals, dependencies, capabilities, and System Economics.

The normal Systems view should answer:

- Is it connected?
- Is it healthy?
- What is it doing?
- Does it need attention?
- What can LEE do with it?

Keep raw contracts, diagnostics, and technical details expandable or under Advanced.

B7. Settings / Advanced

Move developer/admin/internal views under Settings -> Advanced -> System Internals, including as appropriate:

- Identity and Constitution details
- Policies
- raw Event Log
- Engine and Capability Registries
- System Manifest
- Query/provenance diagnostics
- raw governance history
- Self-Test and Self-Improvement
- raw System Economics
- Backups and Restore/Recovery
- scheduler, connector, API, and System Contract diagnostics
- detailed technical health information

Part C - Navigation Consolidation Rules

Unless there is a strong UX reason otherwise, these should no longer be separate primary-sidebar destinations:

- Constitution
- Confidence
- Assumptions
- Decision Impact
- Explain
- Policies
- Decisions
- Waiting
- Evidence
- Imports
- Connectors
- Costs
- Governance
- Backups
- Schedule
- Organization
- Decision Patterns
- Knowledge Map
- Observations
- Strategy
- Simulations
- Reflections
- Learning
- Institutional
- Self-Improvement
- System Economics
- Identity
- Events

Do not delete their functionality. Move it into Work, Knowledge, Systems, contextual detail, or Settings/Advanced.

Contextual information rules

- Confidence appears where it matters - answers, recommendations, projects, interpretations - rather than needing its own destination.
- Governance approvals/holds surface on Today, the affected workflow/project, Systems, and mobile; full history remains available.
- Meaningful events surface in Today, Project Activity, Systems, Timeline, and Ask Lee; the raw Event Log lives in Advanced.
- Self-Improvement surfaces meaningful adaptations only when relevant; raw controls/logs live in Advanced.

Part D - Android Companion

Keep Android simpler than desktop. Target primary navigation:

```
TODAY
ASK LEE
CAPTURE
ALERTS / APPROVALS
SYSTEMS
```

Focus mobile on priorities, quick capture, Ask Lee, waiting/alerts, approvals, system status, and lightweight approved controls. Do not reproduce the complete desktop navigation tree.

Preserve local-first/offline Capture for text, voice, screenshots, photos, links, files, observations, and project updates.

Part E - Visual Design

Preserve the current dark LEE identity, branding, restrained typography, and calm professional operating-console feel.

Reduce sidebar clutter, unnecessary cards, duplicated metrics, oversized healthy-status displays, internal architecture terminology, and visual competition. Use hierarchy, whitespace, tabs, drawers, expandable details, and contextual views.

Non-Negotiable Preservation

This work must preserve:

- all underlying engines and ledgers
- all existing data and database structures
- existing APIs and backend behavior
- current CIL and CerbaSeal connections
- Console functionality
- Android functionality and offline capture
- Windows desktop packaging/build and GitHub release flow
- diagnostics and deep system access under Advanced

Part F - Required Validation and Completion Report

The created tasks are not complete until the relevant checks pass.

CIL authority validation

1. A normal LEE external-model call cannot occur before CIL.
2. T1/T2 CIL resolution prevents an external model call.
3. T3 uses the exact CIL-selected route.
4. Model Router does not override the CIL route.
5. A failed model execution returns to CIL for rerouting.
6. No active duplicate CIL routing/model-selection logic remains in LEE.
7. Model inventory is read-only and does not change routing authority.
8. System Economics records the evidence needed to measure CIL savings.

UI/mobile validation

1. All major capabilities remain reachable after consolidation.
2. Existing data remains intact.
3. Today, Ask Lee, Work, Knowledge, Systems, and Advanced all function.
4. Project consolidated workspace functions.
5. Android navigation functions and offline Capture remains intact.
6. Console typecheck and production build pass.
7. Android typecheck passes.
8. API typecheck/build passes.
9. Desktop build/packaging validation remains valid.
10. Existing behavioral tests continue to pass.

Final report required from Replit

- tasks created and completed
- files changed
- CIL routing authority changes
- duplicate CIL logic removed or disabled
- final AI runtime flow
- model inventory integration status
- CIL economics measurements now available
- old vs new desktop navigation
- old vs new mobile navigation
- pages consolidated
- pages moved to Advanced
- validation/test/build results
- remaining limitations or follow-up tasks

Final target:

SIMPLE TO USE.

EASY TO NAVIGATE.

QUIET WHEN HEALTHY.

INFORMATION APPEARS WHEN IT MATTERS.

CIL OWNS MODEL-ROUTING AUTHORITY.

DEEPLY TECHNICAL DETAIL REMAINS AVAILABLE WHEN NEEDED.

EXTREMELY SOPHISTICATED UNDER THE HOOD.